

Pesticide spray drift in South Africa: Calling for legislation on buffer zones and other measures to protect our people and environment



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Summary

The health and environmental issues associated with pollution caused by agrochemical farming in South Africa go largely undocumented and are mostly disregarded by regulatory authorities.

The extensive use of agrochemical remedies on fruit orchards, vineyards and crop lands, which are cultivated extensively in South Africa, and the associated pesticide spray drift these lead to, is a serious yet under-reported issue. Rural communities are experiencing numerous and severe cases of agrochemical spray drift that affect the environment (accumulations of residues in soil, waterways and underground water that go largely unmonitored), severely affect human health, not only of the farm workers applying the remedies but also communities living adjacent to the sprayed fields. These practices also compromise the livelihood of farmers running agroecological organic operations (either by direct drift or by general contamination of the environment where farming occurs).

Spray drift is an externality associated with chemical farming for which no one is being held accountable. In South Africa, there is a dire lack of research into the causal linkages between this chronic exposure and illness, including non-communicable diseases and reproductive disorders. But there is abundant scientific evidence of such correlation in the international literature (London 2019).

South Africa's legislation on pesticides is regulated by Act 36 of 1947 on agricultural remedies. The 2010 Pest Management Act sought to address the limitations of Act 36 of 1947, with specific consideration given to human health and the environment. However, these provisions have to date failed to materialise in a tangible manner. Overall, the framework regulating pesticide usage in South Africa proves weak and insufficient as it narrows their regulation to spraying protocols and product specification, with still limited consideration given to occupational health, the health of adjacent rural communities, long-term environmental impacts and the need to sanction abuses and compensate victims.

The pesticide industry is highly influential and pressurizes the regulator for the industry's interests to prevail. At the crux of this matter is the fact that South Africa's agricultural system is locked in an agrochemical construct (IPES Food 2016), and that government, the industry and therefore farmers push the flawed notion that food cannot be grown without chemicals.

This paper documents three cases of pesticide drift contamination linked to the cultivation of table grapes, strawberries and blueberries in the Western Cape, and how these affect rural communities. These stories are a small drop in the ocean of spray drift issues plaguing the country. They all give a sense of the disempowerment and hopelessness felt by these local communities, who are virtually left with no recourse.

Given the projected growth of the berry sector in South Africa - a 95% growth in blueberries is for instance projected by 2023, with the mainstay of this growth expected to materialise in the Western Cape (Meintjes 2019) - it can be expected that the issue will only grow more dire in the coming years.

In January 2020, legal correspondence was addressed to the Presidency and several ministries, calling for an overhaul of the agrochemical legislation in South Africa, which it described as 'outdated, fragmented and ineffective'. This paper supports this call and brings further evidence of why this is needed, with a specific focus on the issues related to the spraying of pesticides. The lack of inter-governmental cooperation around the management of pesticides is a critical issue that needs to be addressed as a matter of priority. It is high time the South African Government revised and implemented the 2010 Pest Management Policy to protect the constitutional rights to clean air, clean water and clean food.

This legislation should be revised to set South Africa firmly on the path of pesticide reduction, starting by giving serious priority to non-chemical alternatives. Further measures the Government needs to take specifically to curb this spraying drift issue include: minimizing or prohibiting pesticide use in specific areas, mandating buffer zones to protect non-target rural communities, and safeguard zones where surface and groundwater is used for the abstraction of drinking water, establishing robust surveillance and control systems, putting in place a country-wide ban on aerial spraying, solid provisions for advance warning (of spraying), requiring the use of drift-reducing equipment and setting up sanction and compensation mechanisms so that social and environmental justice can prevail. In addition, as encouraging farmers to transition to softer alternatives is so critical to move towards pesticide reduction in the country, a “transition fund” should be established to support these farmers.

This paper shows how other countries have gone about mandating compulsory buffer zones to preserve people’s health and the environment, especially water bodies. These approaches are not ideal, with buffer distances often being judged as insufficient by civil society, but they are a start. The applicability of the Spray Drift Risk Assessment Tool (SDRAT) to determine the ideal dimension for buffer zones in South Africa is described. The paper also discusses the manner in which the current South African legislation, notably the Pest Management Act, the National Water Act and the Air Quality Act could serve as leverage points to start legislating on buffer zones.

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